Indian Institute of Information Technology Allahabad

Reliability Evaluation of LLM-Based Explanations for Transformer-Detected Satellite Telemetry Anomalies

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Presentation Outline

01 **Background & Motivation**

Why satellite telemetry anomaly detection matters

02 **Problem Statement**

Core research question: LLM explanation reliability

Phase 1 **Data Preprocessing**

ESA benchmark, ZOH resampling, channel normalization + Novelty

Phase 2 **Transformer Architecture & Results**

Anomaly Transformer, dual attention, evaluation metrics

Phase 3 **LLM Explanation Reliability**

AXIS framework — future direction

Ref **References**

Base papers and datasets cited

Background & Motivation

High-Dimensional Telemetry

Modern spacecraft generate continuous streams from sensors, subsystems, and telecommands. Anomalies are rare, short-lived, and often visible only across multiple channels.

Transformer-Based Detection

Attention mechanisms model long-range temporal dependencies and inter-channel relationships. The Anomaly Transformer uses association discrepancy to identify anomalous time points.

The Gap: LLM Explainability

Transformer detectors produce anomaly scores but not explanations. LLMs can generate fluent natural-language explanations — but fluency alone does not guarantee reliability.

Problem Statement & Objectives

Main Research Question

Are LLM-generated explanations for transformer-detected satellite telemetry anomalies reliable?

Transformer detectors produce anomaly scores and predicted labels, but these do not explain why a segment is anomalous.

LLMs can generate fluent explanations — but the central question is whether they are:

- Faithful to the detected anomaly
- Grounded in time-series evidence
- Useful for spacecraft monitoring operators

Objectives

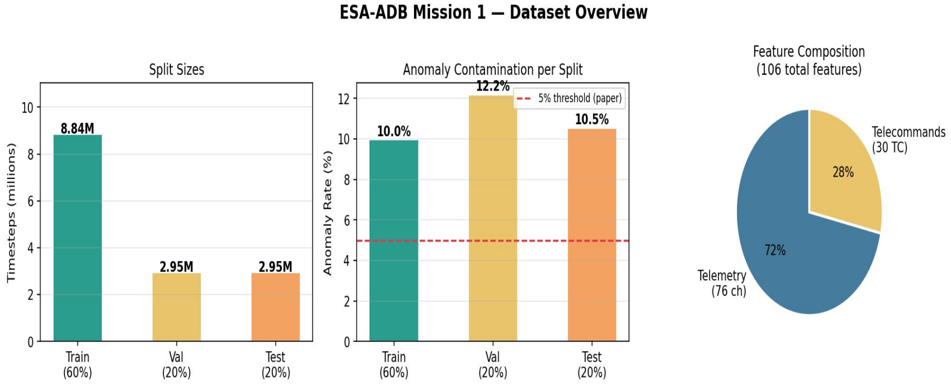
1. Study the ESA satellite telemetry anomaly detection benchmark dataset
2. Preprocess telemetry channels, labels, and telecommands into sliding windows
3. Adapt Anomaly Transformer for prediction on ESA data
4. Address exploding-gradient issue from z-score normalization of telecommands
5. Evaluate model predictions using anomaly scores and classification metrics
6. Use AXIS framework to evaluate LLM explanation reliability (future work)

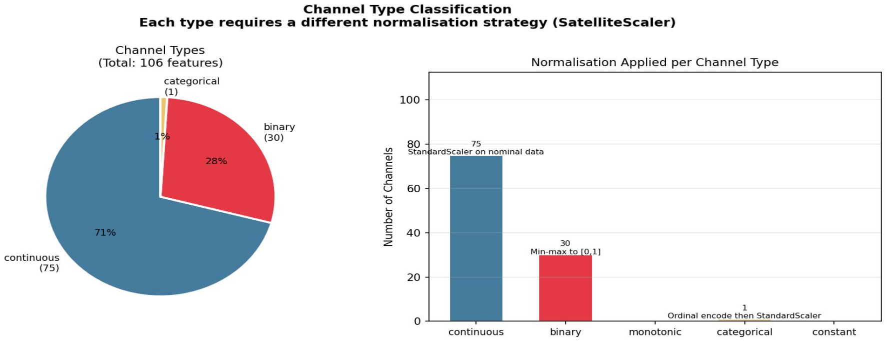
PHASE 1

Data Preprocessing | ESA Benchmark | Novelty: Telecommand Normalization

ESA Benchmark Dataset

Mission:	ESA-ADB Mission 1 — Real satellite telemetry
Channels:	106 total features: 76 telemetry + 30 telecommands
Split:	60% Train (8.84M pts) | 20% Val | 20% Test
Anomaly Rate:	~10–12.2% across splits
Signal Types:	Continuous (71%), Binary (28%), Categorical (1%)





Preprocessing Pipeline

Preprocessing Pipeline — Decision Summary Table

Stage	Decision Made
1. Data Source	ESA-ADB Mission 1
2. Resampling	ZOH @ 30s
3. TC Encoding	Binary impulse @ nearest timestamp
4. Label Build	Logical OR over target channels
5. Split	60 / 20 / 20 chronological
6. Scaler Fit	Nominal train only (label=0)
7. Norm Strategy	Per-type: cont/binary/mono/cat
8. Outlier Fix	np.clip(±10) on load
9. Window	seq_len=128, stride=16
10. Train Filter	Drop anomalous rows (label=1)
11. Total Features	106 (76 telemetry + 30 TC)
12. Train Windows	552,305

Fig 4.1 — Complete Preprocessing Decision Summary Table

Timestamp Alignment & ZOH Resampling

Why ZOH Resampling?

Satellite telemetry channels are sampled at irregular timestamps.

To build multivariate transformer inputs, all channels must be aligned to a common time grid.

Zero-Order Hold (ZOH):
Carries the last known value forward until a new observation arrives — preserving binary signal integrity.

Linear interpolation is NOT used
— it creates artificial intermediate values invalid for binary/quantised signals.

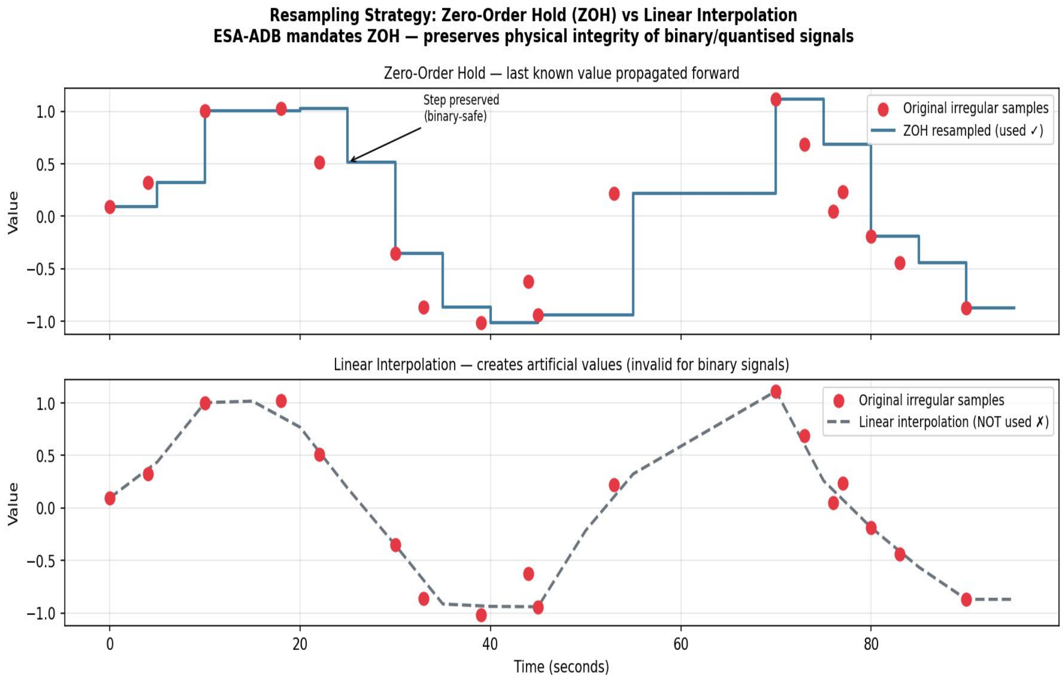


Fig 4.4 — ZOH vs Linear Interpolation on sparse binary telemetry

★ PROBLEM

Telecommand Normalization — Min-Max over Z-Score

Problem: Exploding Gradients

Telecommands are sparse, event-driven signals — mostly zeros with occasional impulses.

Z-score normalization on low-variance channels divides by a very small standard deviation, producing extremely large normalized values.

These large values caused **exploding gradients** and unstable optimization during transformer training.

Solution: Min-Max Normalization

Min-max scaling bounds all telecommand values to [0, 1], regardless of variance:

$$x' = (x - x_{min}) / (x_{max} - x_{min} + \varepsilon)$$

where ε prevents division by zero.

Continuous telemetry channels retain standard z-score normalization. Only the 30 telecommand channels receive this treatment.

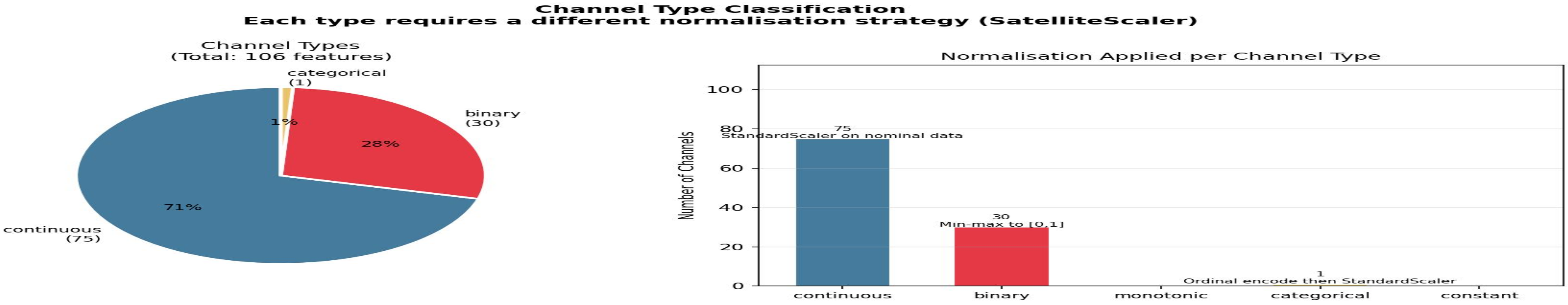


Fig 4.3 — Channel type classification and normalization strategy applied per type

Train/Val/Test Split & Sliding Windows

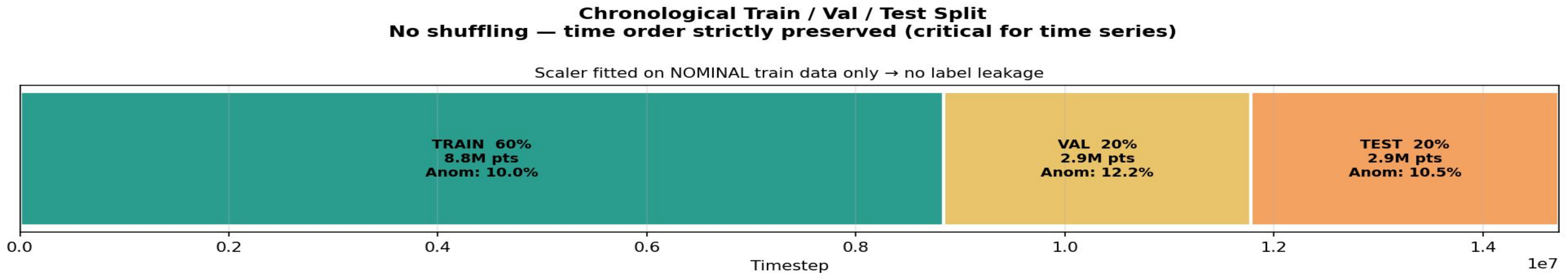


Fig 4.5 — Chronological 60/20/20 split — time order strictly preserved, no shuffling

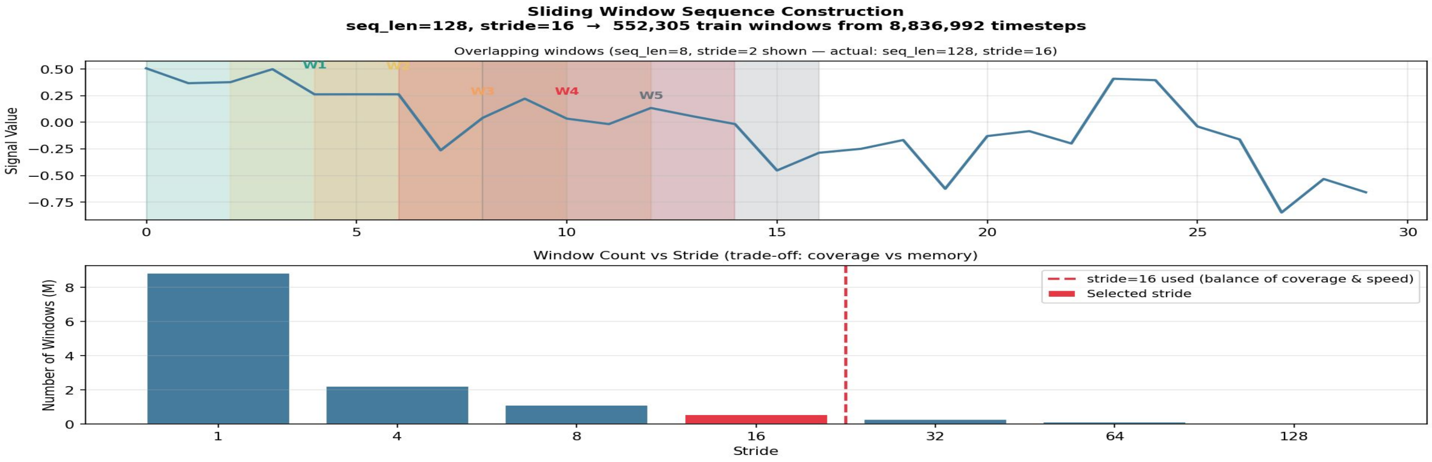


Fig 4.6 — Sliding window construction: seq_len=128, stride=16 → 552,305 training windows

PHASE 2

Anomaly Transformer Architecture | Training | Results

Anomaly Transformer Architecture

Core Idea: Normal time points build broad temporal associations; anomalous points concentrate associations on nearby neighbors. This difference — association discrepancy — drives detection.

Series Association S

Learned directly from input via scaled dot-product attention:
$S = \mathrm{Softmax}(QK^T / \sqrt{d\_model})$
Captures actual temporal relationships in the data.

Prior Association P

Learnable Gaussian kernel encoding adjacent-concentration bias:
$P\_ij = \mathrm{Rescale}(\ \exp(-|j-i|^2 / 2\sigma_i^2)\ )$
σ_i adapts the local neighborhood width per time point.

Association Discrepancy

Symmetrized KL divergence between P and S across all layers:
$\mathrm{AssDis} = (1/L)\ \Sigma\ [\ KL(P^l \| S^l) + KL(S^l \| P^l)\ ]$
Anomalous points show smaller discrepancy.

Anomaly Score: *$\mathrm{AnomalyScore}(X) = \mathrm{Softmax}(-\mathrm{AssDis}(P,S;X)) \odot \|X_i - \hat{X}_i\|^2$ → Combines attention discrepancy + reconstruction error*

Minimax Learning & Model Training

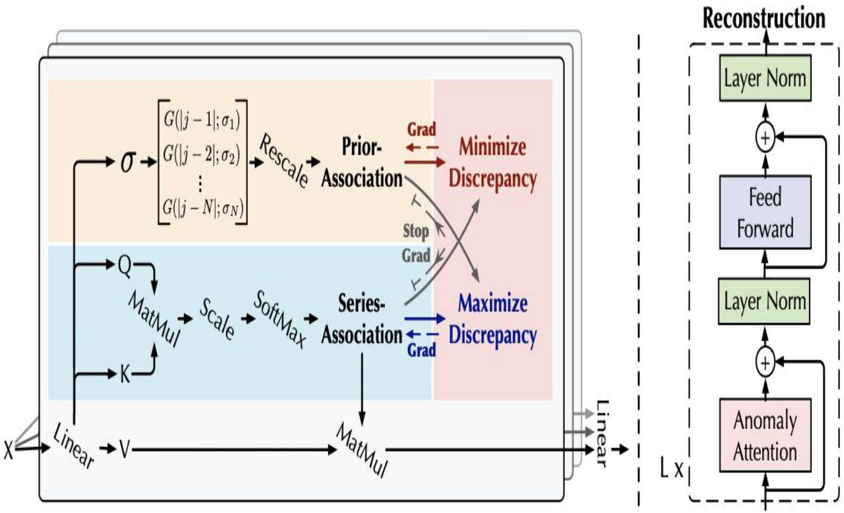
Minimax Association Learning

Total Loss:
L = ‖X − X̂‖²_F − λ‖AssDis(P,S;X)‖₁

Minimization phase:
L_min: Prior P approximates detached series S.

Maximization phase:
L_max: Series S enlarges discrepancy while preserving reconstruction quality.

Why it works:
Prevents the Gaussian prior from collapsing while forcing learned series association to expose normal vs anomalous differences.



Results: Anomaly Score Distribution & Confusion Matrix

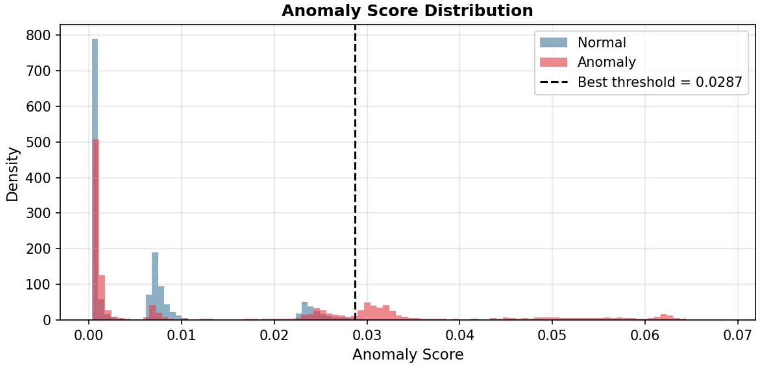


Fig 4.7 — Anomaly Score Distribution (best threshold = 0.0287)

Precision = 62%
Recall =33.78%
F-1 Score = 43.74%

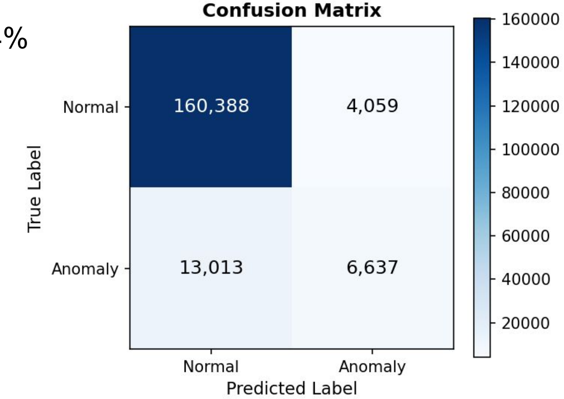


Fig 4.8 — Confusion Matrix after thresholding

True Normal

160,388

Correctly identified normal windows

True Anomaly

6,637

Correctly detected anomaly windows

False Negative

13,013

Missed anomalies — high recall gap

False Positive

4,059

Normal windows flagged as anomaly

Results: Threshold Sensitivity & Association Behaviour

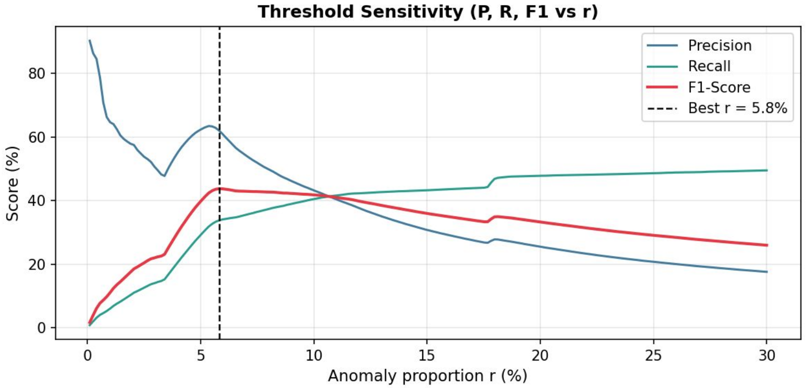


Fig 4.9 — Precision, Recall & F1 vs anomaly proportion threshold (best r ≈ 5.8%)

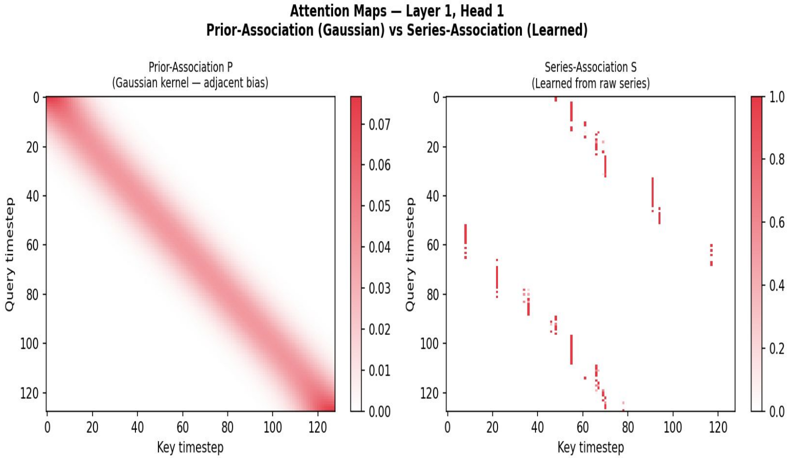


Fig 4.10 — Prior Association (Gaussian) vs Series Association (Learned) — Layer 1, Head 1

Threshold Insight

Precision drops and recall rises as threshold lowers. The optimal point at r ≈ 5.8% maximizes F1-score. The model is sensitive to threshold choice — highlighting importance of careful threshold selection for real deployment.

Association Discrepancy

The Gaussian prior association (left) shows smooth diagonal concentration. The learned series association (right) exhibits sparse, irregular attention — confirming the model captures anomaly-driven temporal patterns distinct from normal behavior.

PHASE 3

LLM Explanation Reliability | AXIS Framework | Future Direction

LLM Explanation Reliability: AXIS Framework

Transformer anomaly detectors produce scores — not explanations. LLMs can generate fluent natural-language descriptions, but fluency ≠ reliability. Are the explanations faithful, grounded, and useful?

1

Anomaly Window

Detected anomalous time-series segment from transformer output, with nearby normal context for comparison.

2

Channel Context

Relevant telemetry channel metadata, anomaly score, and key statistical summaries passed to the LLM.

3

LLM Explanation

Large language model generates natural-language explanation of why the segment was flagged as anomalous.

4

AXIS Reliability Eval

Evaluate factual correctness, consistency with anomaly window, operator clarity, and faithfulness to model output.

Reliability Criteria: Factual Correctness • Consistency with Anomaly Window • Grounded in Time-Series Evidence • Clarity for Human Operators • Faithfulness to Model Output

Conclusion

✓ Base Paper Implemented

Anomaly Transformer adapted and trained on ESA-ADB Mission 1 telemetry benchmark.

★ Novelty: Telecommand Normalization

Z-score normalization caused exploding gradients on sparse TC channels; min-max scaling resolved training instability.

✓ Pipeline Complete

End-to-end: ESA loading → ZOH resampling → sliding windows → transformer training → anomaly score evaluation.

→ Future: AXIS Explanation Reliability

Detected anomaly windows + channel context will feed into AXIS framework to test LLM explanation faithfulness.

References

[1]

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European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry.

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AXIS: Explainable Time Series Anomaly Detection with Large Language Models.

arXiv preprint, 2025.

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Anomaly Transformer: Time Series Anomaly Detection with Association Discrepancy.

International Conference on Learning Representations (ICLR), 2022.

Dataset: ESA-ADB available at ESA Φ-lab. Code implemented in Python using PyTorch.